# MJD122 (NPN)
# MJD127 (PNP)

**Preferred Device**

# Complementary Darlington Power Transistor

## DPAK For Surface Mount Applications

Designed for general purpose amplifier and low speed switching applications.

### Features

- Lead Formed for Surface Mount Applications in Plastic Sleeves
- Surface Mount Replacements for 2N6040–2N6045 Series, TIP120–TIP122 Series, and TIP125–TIP127 Series
- Monolithic Construction With Built–in Base–Emitter Shunt Resistors
- High DC Current Gain: $h_{FE}$ = 2500 (Typ) @ $I_C$ = 4.0 Adc
- Epoxy Meets UL 94 V–0 @ 0.125 in
- ESD Ratings: Human Body Model, 3B > 8000 V
  Machine Model, C > 400 V
- Pb–Free Packages are Available

### MAXIMUM RATINGS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Collector–Emitter Voltage | $V_{CEO}$ | 100 | Vdc |
| Collector–Base Voltage | $V_{CB}$ | 100 | Vdc |
| Emitter–Base Voltage | $V_{EB}$ | 5 | Vdc |
| Collector Current – Continuous<br>– Peak | $I_C$ | 8<br>16 | Adc |
| Base Current | $I_B$ | 120 | mAdc |
| Total Power Dissipation @ $T_C$ = 25°C<br>Derate above 25°C | $P_D$ | 20<br>0.16 | W<br>W/°C |
| Total Power Dissipation (Note 1)<br>@ $T_A$ = 25°C<br>Derate above 25°C | $P_D$ | <br>1.75<br>0.014 | <br>W<br>W/°C |
| Operating and Storage Junction Temperature Range | $T_J$, $T_{stg}$ | –65 to +150 | °C |

### THERMAL CHARACTERISTICS

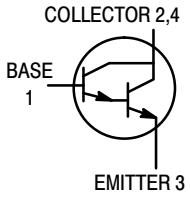
| Characteristic | Symbol | Max | Unit |
|---|---|---|---|
| Thermal Resistance Junction–to–Case | $R_{\theta JC}$ | 6.25 | °C/W |
| Thermal Resistance Junction–to–Ambient (Note1) | $R_{\theta JA}$ | 71.4 | °C/W |

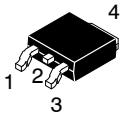
Maximum ratings are those values beyond which device damage can occur. Maximum ratings applied to the device are individual stress limit values (not normal operating conditions) and are not valid simultaneously. If these limits are exceeded, device functional operation is not implied, damage may occur and reliability may be affected.

1. These ratings are applicable when surface mounted on the minimum pad sizes recommended.

**ON Semiconductor®**

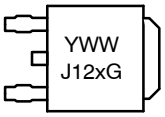
**SILICON**
**POWER TRANSISTOR**
**8 AMPERES**
**100 VOLTS, 20 WATTS**




**DPAK**
**CASE 369C**
**STYLE 1**

**MARKING DIAGRAM**




Y = Year
WW = Work Week
x = 2 or 7
G = Pb–Free Package

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| MJD122 | DPAK | 75 Units/Rail |
| MJD122G | DPAK (Pb–Free) | 75 Units/Rail |
| MJD122T4 | DPAK | 2500/Tape & Reel |
| MJD122T4G | DPAK (Pb–Free) | 2500/Tape & Reel |
| MJD127 | DPAK | 75 Units/Rail |
| MJD127G | DPAK (Pb–Free) | 75 Units/Rail |
| MJD127T4 | DPAK | 2500/Tape & Reel |
| MJD127T4G | DPAK (Pb–Free) | 2500/Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

**Preferred** devices are recommended choices for future use and best overall value.

**ELECTRICAL CHARACTERISTICS** ($T_C$ = 25°C unless otherwise noted)

| Characteristic | Symbol | Min | Max | Unit |
|---|---|---|---|---|
| **OFF CHARACTERISTICS** | | | | |
| Collector−Emitter Sustaining Voltage ($I_C$ = 30 mAdc, $I_B$ = 0) | $V_{CEO(sus)}$ | 100 | − | Vdc |
| Collector Cutoff Current ($V_{CE}$ = 50 Vdc, $I_B$ = 0) | $I_{CEO}$ | − | 10 | μAdc |
| Collector Cutoff Current ($V_{CB}$ = 100 Vdc, $I_E$ = 0) | $I_{CBO}$ | − | 10 | μAdc |
| Emitter Cutoff Current ($V_{BE}$ = 5 Vdc, $I_C$ = 0) | $I_{EBO}$ | − | 2 | mAdc |
| **ON CHARACTERISTICS** | | | | |
| DC Current Gain<br>($I_C$ = 4 Adc, $V_{CE}$ = 4 Vdc)<br>($I_C$ = 8 Adc, $V_{CE}$ = 4 Vdc) | $h_{FE}$ | <br>1000<br>100 | <br>12,000<br>− | − |
| Collector−Emitter Saturation Voltage<br>($I_C$ = 4 Adc, $I_B$ = 16 mAdc)<br>($I_C$ = 8 Adc, $I_B$ = 80 mAdc) | $V_{CE(sat)}$ | <br>−<br>− | <br>2<br>4 | Vdc |
| Base−Emitter Saturation Voltage (Note 2) ($I_C$ = 8 Adc, $I_B$ = 80 mAdc) | $V_{BE(sat)}$ | − | 4.5 | Vdc |
| Base−Emitter On Voltage ($I_C$ = 4 Adc, $V_{CE}$ = 4 Vdc) | $V_{BE(on)}$ | − | 2.8 | Vdc |
| **DYNAMIC CHARACTERISTICS** | | | | |
| Current−Gain−Bandwidth Product ($I_C$ = 3 Adc, $V_{CE}$ = 4 Vdc, f = 1 MHz) | $\|h_{fe}\|$ | 4 | − | MHz |
| Output Capacitance ($V_{CB}$ = 10 Vdc, $I_E$ = 0, f = 0.1 MHz)<br>MJD127<br>MJD122 | $C_{ob}$ | <br>−<br>− | <br>300<br>200 | pF |
| Small−Signal Current Gain ($I_C$ = 3 Adc, $V_{CE}$ = 4 Vdc, f = 1 kHz) | $h_{fe}$ | 300 | − | − |

2. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2%.

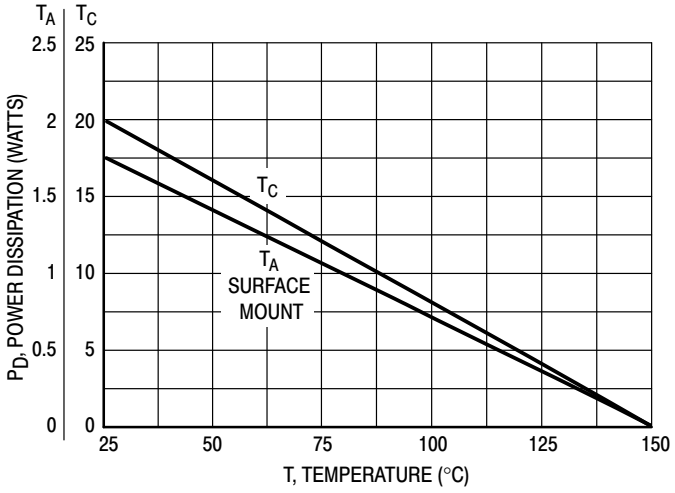
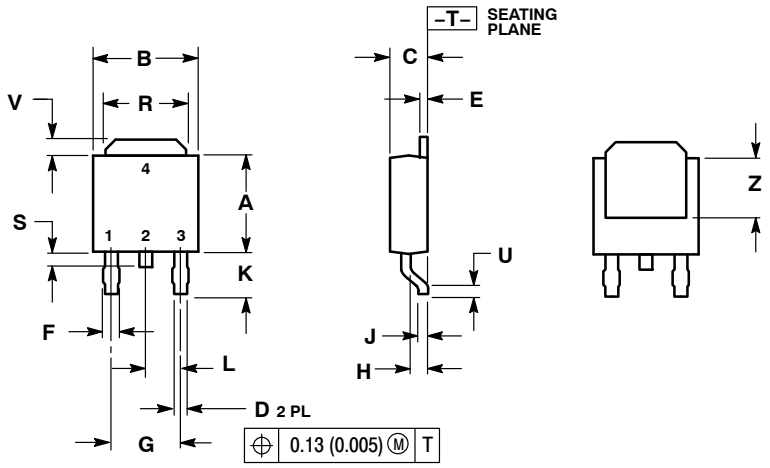



**Figure 1. Power Derating**

## PACKAGE DIMENSIONS

**DPAK**
CASE 369C
ISSUE O

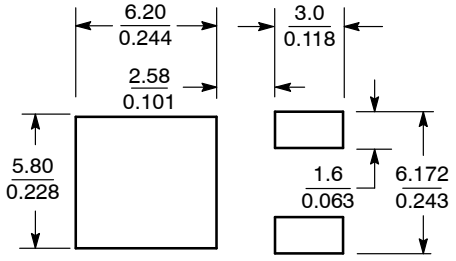



NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.

| | INCHES | | MILLIMETERS | |
|---|---|---|---|---|
| DIM | MIN | MAX | MIN | MAX |
| A | 0.235 | 0.245 | 5.97 | 6.22 |
| B | 0.250 | 0.265 | 6.35 | 6.73 |
| C | 0.086 | 0.094 | 2.19 | 2.38 |
| D | 0.027 | 0.035 | 0.69 | 0.88 |
| E | 0.018 | 0.023 | 0.46 | 0.58 |
| F | 0.037 | 0.045 | 0.94 | 1.14 |
| G | 0.180 BSC | | 4.58 BSC | |
| H | 0.034 | 0.040 | 0.87 | 1.01 |
| J | 0.018 | 0.023 | 0.46 | 0.58 |
| K | 0.102 | 0.114 | 2.60 | 2.89 |
| L | 0.090 BSC | | 2.29 BSC | |
| R | 0.180 | 0.215 | 4.57 | 5.45 |
| S | 0.025 | 0.040 | 0.63 | 1.01 |
| U | 0.020 | --- | 0.51 | --- |
| V | 0.035 | 0.050 | 0.89 | 1.27 |
| Z | 0.155 | --- | 3.93 | --- |

STYLE 1:
PIN 1. BASE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

## SOLDERING FOOTPRINT*




SCALE 3:1 (mm / inches)

*For additional information on our Pb–Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.